SQL Quick Reference

Part 2: Retrieving Data

Each command below has a short explanation and an example. The examples use a students table with the columns id, name, age, major, credits and email.

1. SELECT ... FROM

Retrieves data from a table. List the columns you want after SELECT, or use a star (*) for every column.

Example:

```
SELECT name, major
FROM students;
```

2. WHERE

Keeps only the rows that meet a condition. Conditions use comparison operators such as = (equal to), < (less than), > (greater than) and <> (not equal to).

Example:

```
SELECT name
FROM students
WHERE age >= 20;
```

3. AND, OR, NOT

AND needs both conditions to be true, and OR needs at least one. NOT reverses a condition.

Example:

```
SELECT name
FROM students
WHERE major = 'Biology' AND age < 20;
```

4. DISTINCT

Removes duplicate values from the result, so each value appears only once.

Example:

```
SELECT DISTINCT major
FROM students;
```

5. ORDER BY

Sorts the result from smallest to largest, or A to Z. Add DESC to sort from largest to smallest.

Example:

```
SELECT name, age
FROM students
ORDER BY age DESC;
```

6. LIKE

Finds values that match a pattern. The percent sign (%) stands for any number of characters.

Example:

```
SELECT name
FROM students
WHERE name LIKE 'M%';
```

7. IN

Checks whether a value is in a list. It is a shorter way to write several OR conditions.

Example:

```
SELECT name
FROM students
WHERE major IN ('Biology', 'History');
```

8. IS NULL

Finds rows where a column is empty. Always use IS NULL, because = NULL never matches anything; IS NOT NULL finds rows that do have a value.

Example:

```
SELECT name
FROM students
WHERE email IS NULL;
```

9. COUNT, SUM, AVG, MIN, MAX

Calculate one value from many rows: COUNT counts rows, SUM adds, AVG averages, and MIN and MAX find the smallest and largest values.

Example:

```
SELECT COUNT(*), SUM(credits), AVG(age), MIN(age), MAX(age)
FROM students;
```